Why Machine Learning?
Types of Learning & Machine Learning Workflow

Why Do We Need Machine Learning?

Traditional Programming has limitations

- Difficult to write rules for every problem.
- Rules become complex as data grows.
- Cannot easily adapt to new situations.

Machine Learning

- Learns patterns automatically from data.
- Improves performance with experience.
- Handles large and complex datasets.
- Example:
  Email Spam Detection

Traditional Programming vs Machine Learning

Traditional Programming:

Input + Rules → Output

Machine Learning:

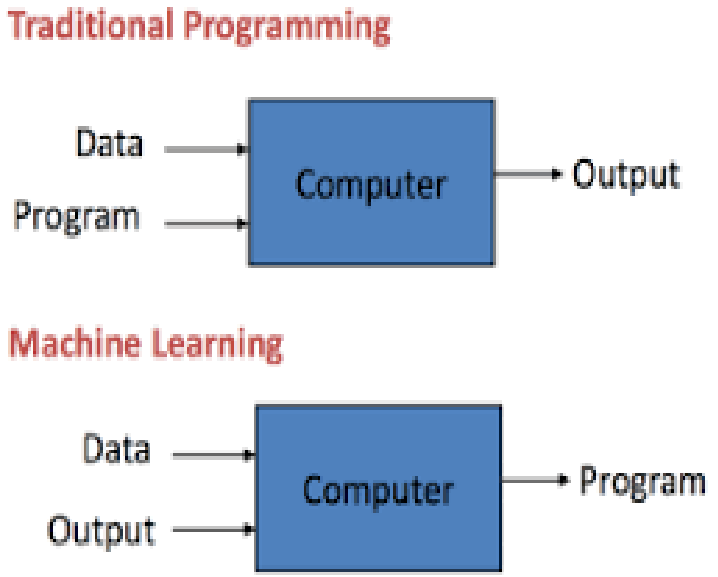
- Input + Output → Machine Learns Rules → Prediction
- Example: House Price Prediction

Machine Learning

Arthur Samuel, a pioneer in the field of artificial intelligence and computer gaming, coined the term **"Machine Learning"** as – **"Field of study that gives computers the capability to learn without being explicitly programmed"**.

How it is different from traditional Programming:

- In Traditional Programming, we feed the Input, Program logic and run the program to get output.
- In Machine Learning, we feed the input, output and run it on machine during training and the machine creates its own logic, which is being evaluated while testing.



What is Learning?

A computer improves its performance by learning from past data without being explicitly programmed.

Example:
A child learns to identify fruits after seeing many examples.

Similarly,
A machine learns by analyzing many examples.

Terminologies that one should know before starting Machine Learning:

- **Model:** A model is a **specific representation** learned from data by applying some machine learning algorithm. A model is also called **hypothesis**.

- **Feature:** A feature is an individual measurable property of our data. A set of numeric features can be conveniently described by a **feature vector**. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, **etc.**

- **Target(Label):** A target variable or label is the value to be predicted by our model. For the fruit example discussed in the features section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training:** The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.

- **Prediction:** Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

Types of Machine Learning

There are three major types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

The model learns from labelled data.

Label = Correct Answer

Examples:

- Predict house price
- Predict student pass/fail
- Disease prediction
- Email spam detection

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree
- KNN
- Naïve Bayes
- SVM

Supervised Learning Diagram

Training Data

Input + Output Labels

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Machine Learning Algorithm

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Trained Model

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Prediction

Types of Learning

- **Supervised Learning**
- **Unsupervised Learning**
- **Semi-Supervised Learning**

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1. **Supervised Learning**: Supervised learning is when the model is getting trained on a labelled dataset. **Labelled** dataset is one which have both input and output parameters. In this type of learning both training and validation datasets are labelled as shown in the figures below.

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0

Classification

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987

Regression

Types of Supervised Learning:

- **Classification**
- **Regression**

Classification : It is a Supervised Learning task where output is having defined labels(discrete value). For example in above Figure A, Output – Purchased has defined labels i.e. 0 or 1 ; 1 means the customer will purchase and 0 means that customer won't purchase. It can be either binary or multi class classification. In **binary** classification, model predicts either 0 or 1 ; yes or no but in case of **multi class** classification, model predicts more than one class.

Example: Gmail classifies mails in more than one classes like social, promotions, updates, offers.

Regression : It is a Supervised Learning task where output is having continuous value. Example in before regression Figure, Output – Wind Speed is not having any discrete value but is continuous in the particular range. The goal here is to predict a value as much closer to actual output value as our model can and then evaluation is done by calculating error value. The smaller the error the greater the accuracy of our regression model.

Example of Supervised Learning Algorithms:

- ❑ Linear Regression
- ❑ Nearest Neighbor
- ❑ Gaussian Naive Bayes
- ❑ Decision Trees
- ❑ Support Vector Machine (SVM)
- ❑ Random Forest

Unsupervised Learning:

Unsupervised learning is the training of machine using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance. Here the task of machine is to group unsorted information according to similarities, patterns and differences without any prior training of data. Unsupervised machine learning is more challenging than supervised learning due to the absence of labels.

Types of Unsupervised Learning:

- **Clustering**
- **Association**

Unsupervised Learning Diagram

- Input Data
- ↓
- Find Hidden Patterns
- ↓
- Clusters

Unsupervised Learning

Definition:
The model learns patterns from unlabelled data.

Examples:

- Customer segmentation
- Product grouping
- Market basket analysis
- Image clustering

Algorithms:

- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA

Unsupervised Learning Diagram

- Input Data
- ↓
- Find Hidden Patterns
- ↓
- Clusters

Clustering: A clustering problem is where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behavior.

Association: An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.

Examples of unsupervised learning algorithms are:

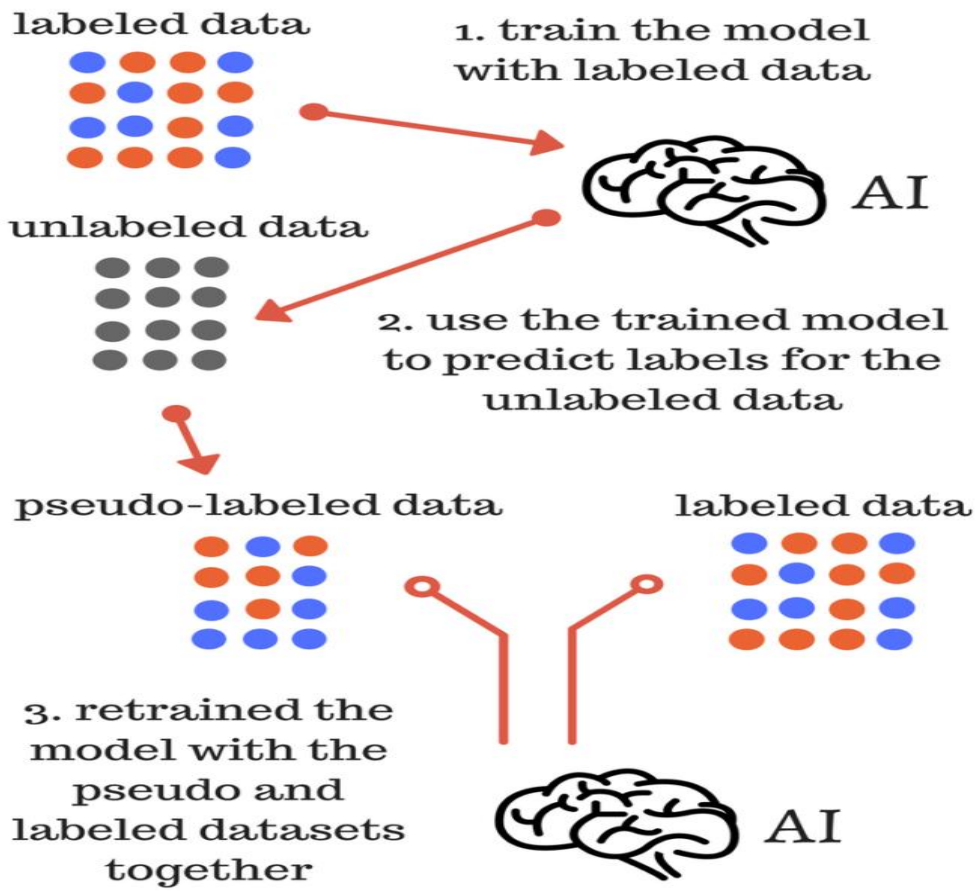
- ❑ k-means for clustering problems.
- ❑ Apriori algorithm for association rule learning problems

The most basic disadvantage of any **Supervised Learning** algorithm is that the dataset has to be hand-labeled either by a Machine Learning Engineer or a Data Scientist. This is a very *costly process*, especially when dealing with large volumes of data. The most basic disadvantage of any **Unsupervised Learning** is that it's **application spectrum is limited**.

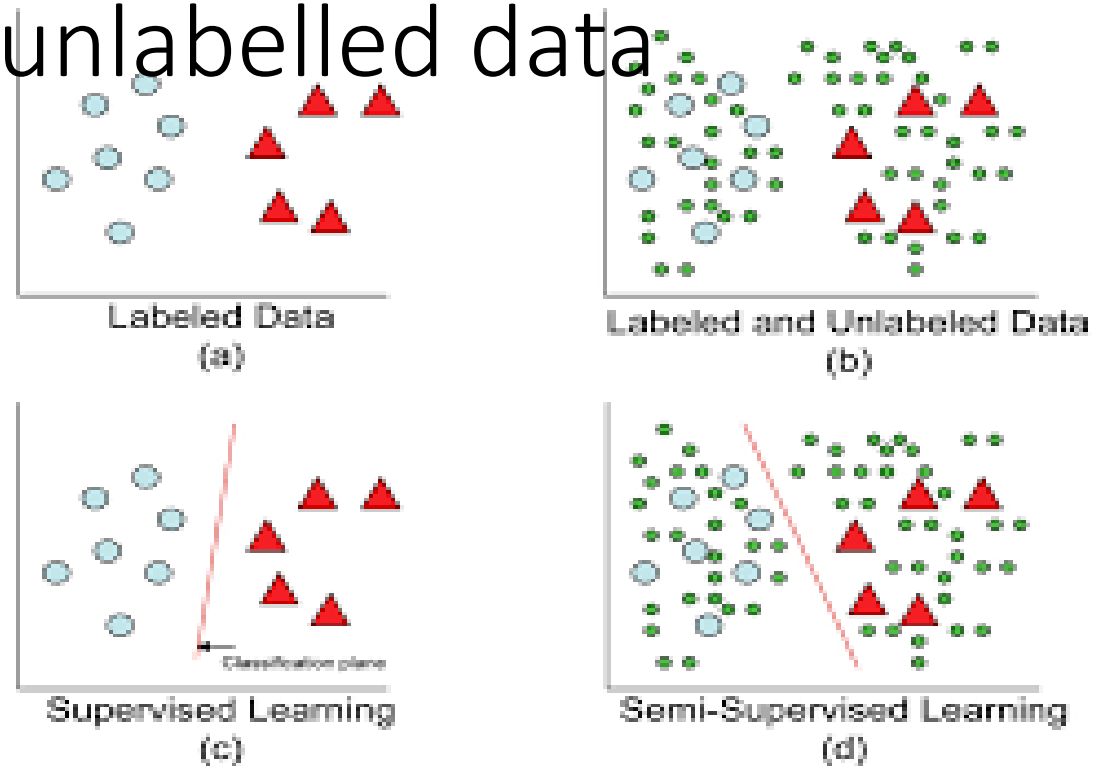
Semi-supervised machine learning:

To counter these disadvantages, the concept of **Semi-Supervised Learning** was introduced. In this type of learning, the algorithm is trained upon a combination of labeled and unlabeled data. Typically, this combination will contain a very small amount of labeled data and a very large amount of unlabeled data.

- In semi supervised learning labelled data is used to learn a model and using that model unlabeled data is labelled called pseudo labelling now using whole data model is trained for further use



Model with labellled data and model with both labelled and unlabelled data



Intuitively, one may imagine the three types of learning algorithms as Supervised learning where a student is under the supervision of a teacher at both home and school, Unsupervised learning where a student has to figure out a concept himself and Semi-Supervised learning where a teacher teaches a few concepts in class and gives questions as homework which are based on similar concepts.

REGRESSION

Regression is a statistical measurement used in finance, investing, and other disciplines that attempts to determine the strength of the relationship between one dependent variable and a series of other changing variables or independent variable

Types of regression

- linear regression
  - Simple linear regression
  - Multiple linear regression
- Polynomial regression
- Decision tree regression
- Random forest regression

Reinforcement Learning

The model learns by interacting with the environment through rewards and penalties.

Examples:

- Robot navigation
- Self-driving cars
- Chess playing
- Game AI

Reinforcement Learning Diagram

Agent

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Action

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Environment

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Reward / Penalty

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Learning